

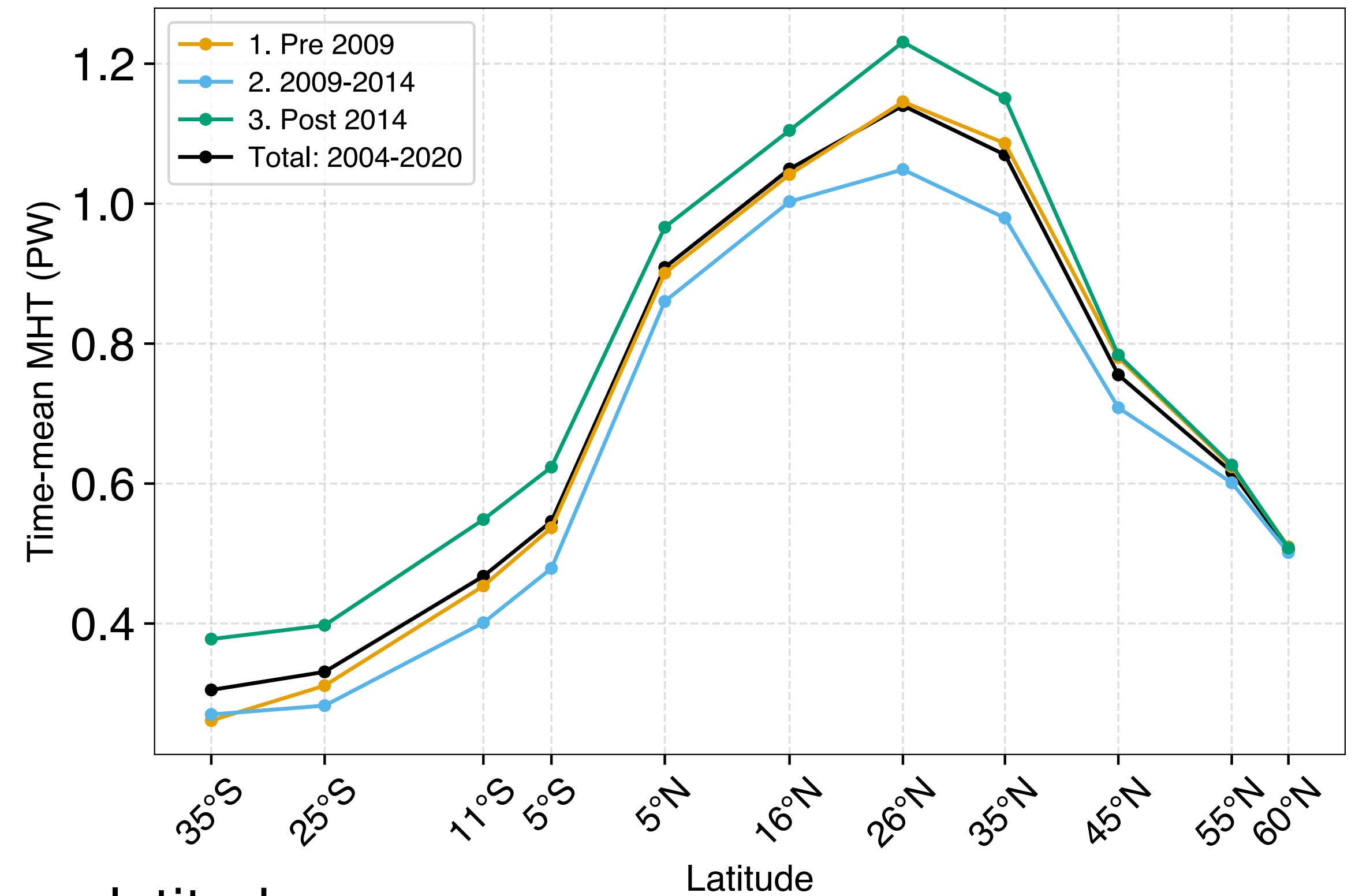
Spatiotemporal coherence of MHT in the Atlantic Ocean

4. Closing the heat budget at 16° - 5° N, connecting with NAO index

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Time-mean MHT for 3 periods

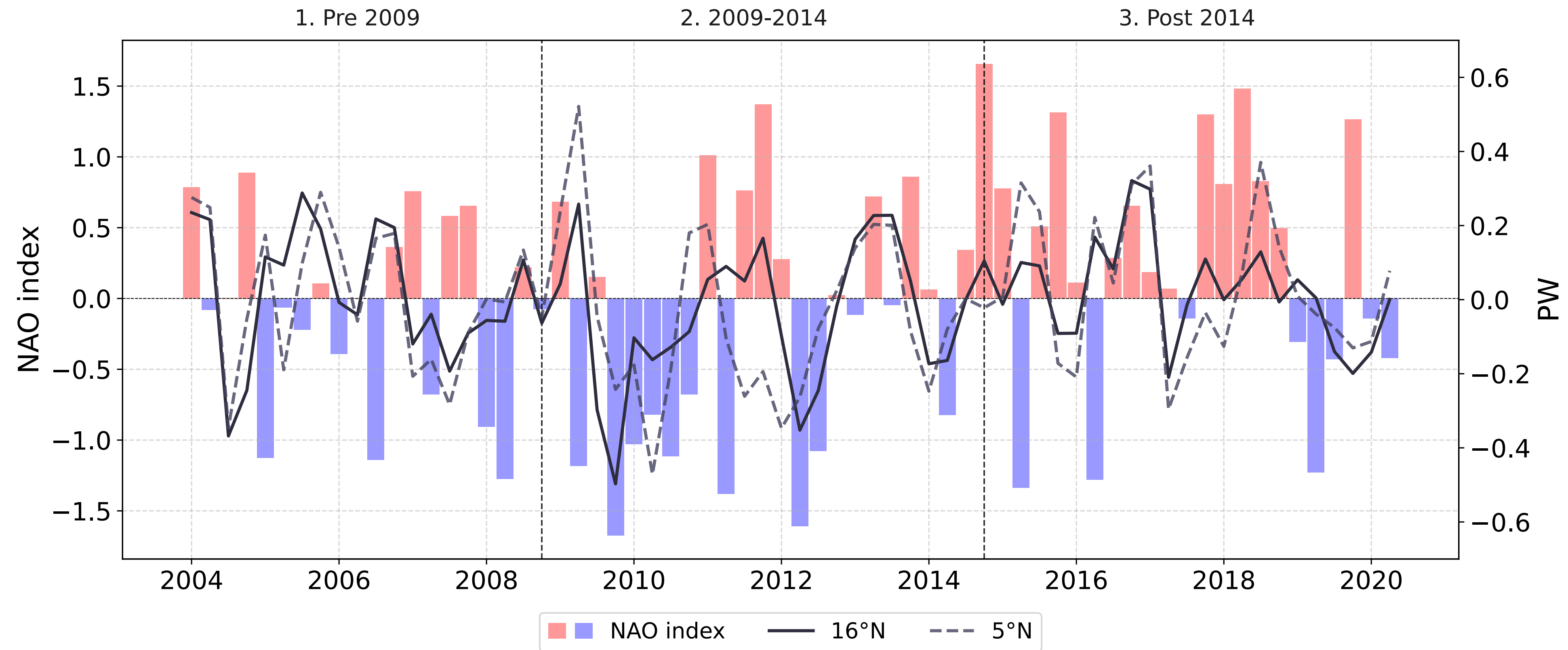
Period	16°N MHT (PW)	5°N MHT (PW)
Pre 2009	1.04	0.90
2009-2014	1.00	0.86
Post 2014	1.10	0.97



- Time mean MHT is lower in 2009-2014 period across latitudes
- Post 2014 is higher than state before coherence break
- Changes by similar amount at the two break latitudes

NAO & MHT anomalies

How do different latitudes (specifically the break latitudes) connect to the NAO index, for the different periods?



MHT and NAO correlation

- For period of negative NAO starting with the anomalous event in 2009 the heat transport in the subtropical gyre shows a higher correlation with NAO index (~0.5)
- Not visible for the other 2 periods

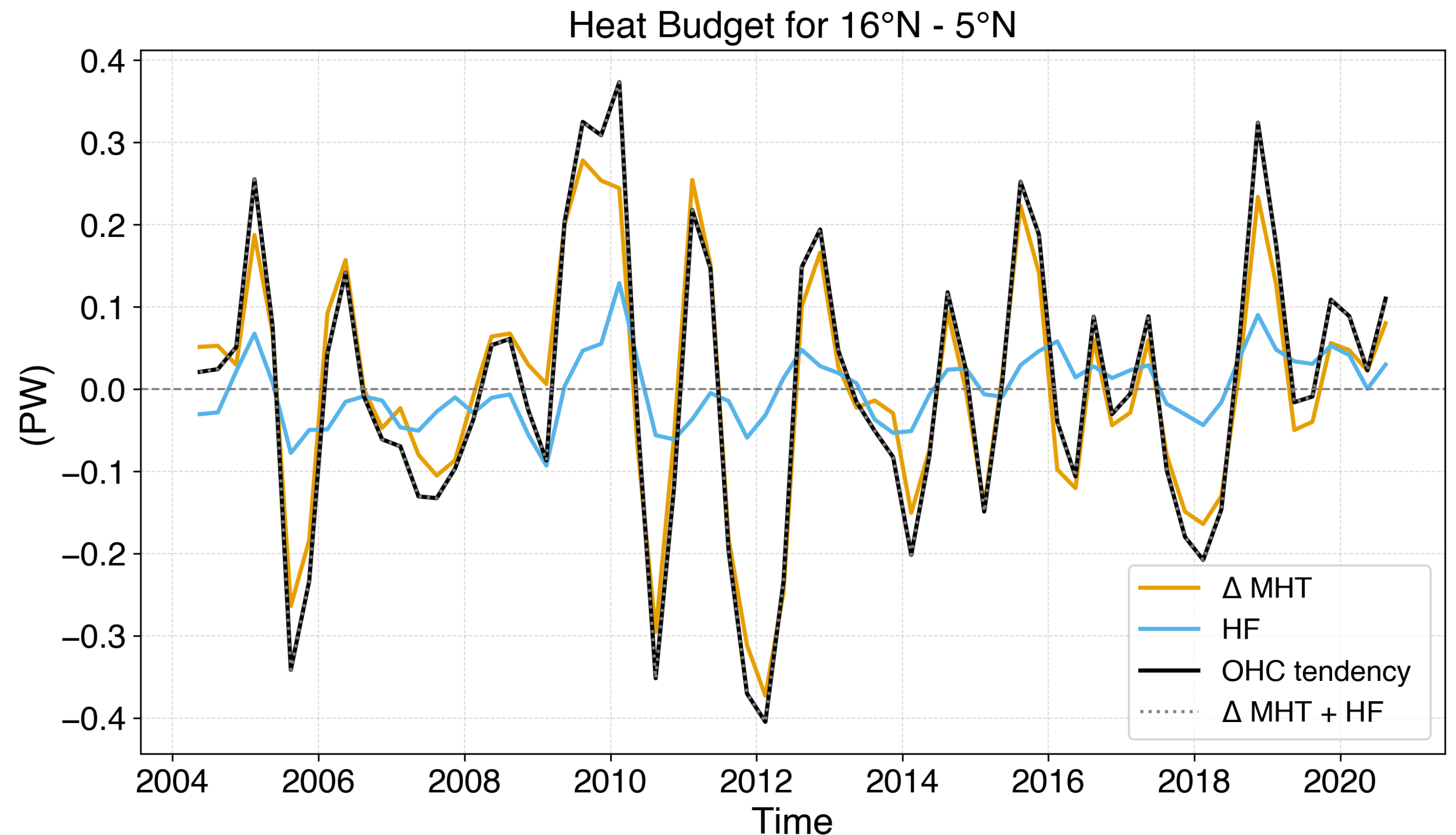
⇒ In period of lower over MHT, MHT anomalies in ST gyre region connect slightly with NAO index ?

Period	35°N	26°N	16°N	5°N
Pre 2009	-0.18	-0.31	-0.20	-0.17
2009-2014	0.52	0.53	0.51	0.16
Post 2014	0.21	0.12	-0.01	-0.15

Correlation of MHT at latitudes with NAO index for each period, blue shaded values are significant (95%)

Heat budget

- How can I explain the coherence break happening in 2009-2014 at 16°- 5°N?
- How much of the change in heat is explained through air sea heat flux changes?



Reduction compared to pre 2009 baseline

16°N - 5°N :

Pre 2009:

$$\text{OHCT} = \text{HTC} + \text{HF} = (MHT_{in} - MHT_{out}) + \text{HF} = -0.141 + 0.127 = \mathbf{-0.014 \text{ PW}}$$

→ mild cooling of the region (more heat transport northwards than inward and uptake)

2009-2014:

$$\text{OHCT} = -0.143 + 0.146 = \mathbf{+0.003 \text{ PW}}$$

→ **mild warming** (~same MHT, but increased heat uptake, changing signs for OHCT)

2009-2014 - baseline(pre2009):

$$\Delta\text{OHCT} = \mathbf{+0.017 \text{ PW}}$$

$$\Delta\text{HF} = +0.019 \text{ PW} \rightarrow \mathbf{\text{Heat flux uptake is 15\% higher in 2nd period}}$$

$$\Delta\text{HTC} = -0.002 \text{ PW}$$

⇒ increase in OHCT is **only caused by increased uptake of heat** from atmosphere, not by a change in transport convergence inside the region

Reduction compared to pre 2009 baseline

16°N

Pre 2009:

1.042 PW

2009-2014:

1.003 PW

2009-2014 - baseline(pre2009):

-3.7% reduced MHT

5°N

Pre 2009:

0.901 PW

2009-2014:

0.860 PW

2009-2014 - baseline(pre2009):

-4.5 % reduced MHT

⇒ **circulation related reduction in MHT** across region ⇒ broader circulation change

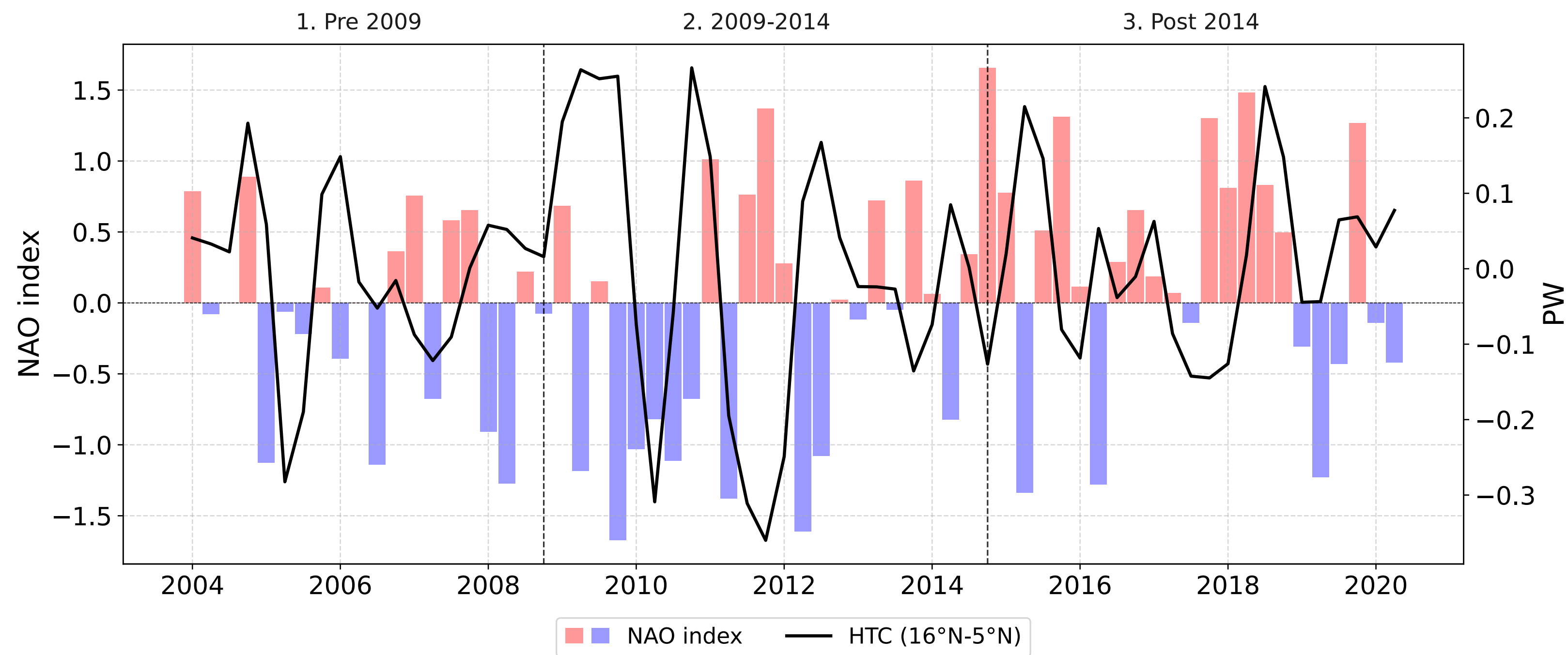
However decline is almost uniform at 5°N and 16°N, which connects with almost no difference in HTC inside the region.

region leading to increase in OHCT

Heat budget

- From literature it is known that in the time period of negative NAO event (2009/2010) there is a reduction of 2.5 Sv in AMOC from 2009-2016 (*Bryden et al. 2020*)
- State of reduced MHT across latitudes is probably caused by this decrease of overturning strength
- In such a state of reduced overturning, there is said to be more recirculation (*McCarthy 2012*)
 - heat is rather recirculated within the gyres
 - This connects to the coherence splitting between 16°N and 5°N since the subtropical gyres southern boundary is roughly in this region
 - Change in MHT is likely coming from circulation, since there is even an increase in heat uptake visible in the region 16°N-5°N
 - Correlation with NAO in this period hints towards circulation driven as well, as the „faster“ impact would be through wind & Ekman circulation changes

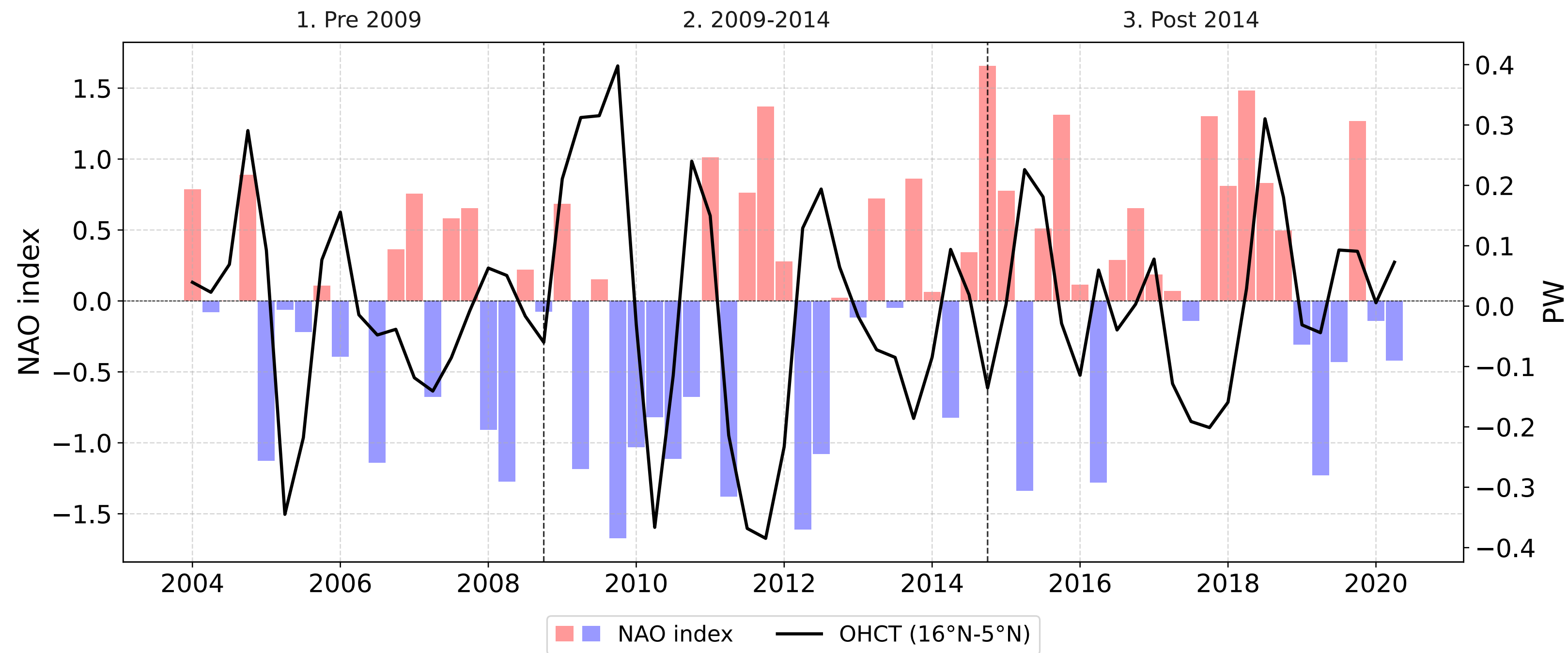
NAO & HTC anomalies



Period	16°N-5°N
Pre 2009	0.03
2009-2014	-0.33
Post 2014	-0.25

Correlation of OHCT in region with NAO index for each period, blue shaded values are significant (95%)

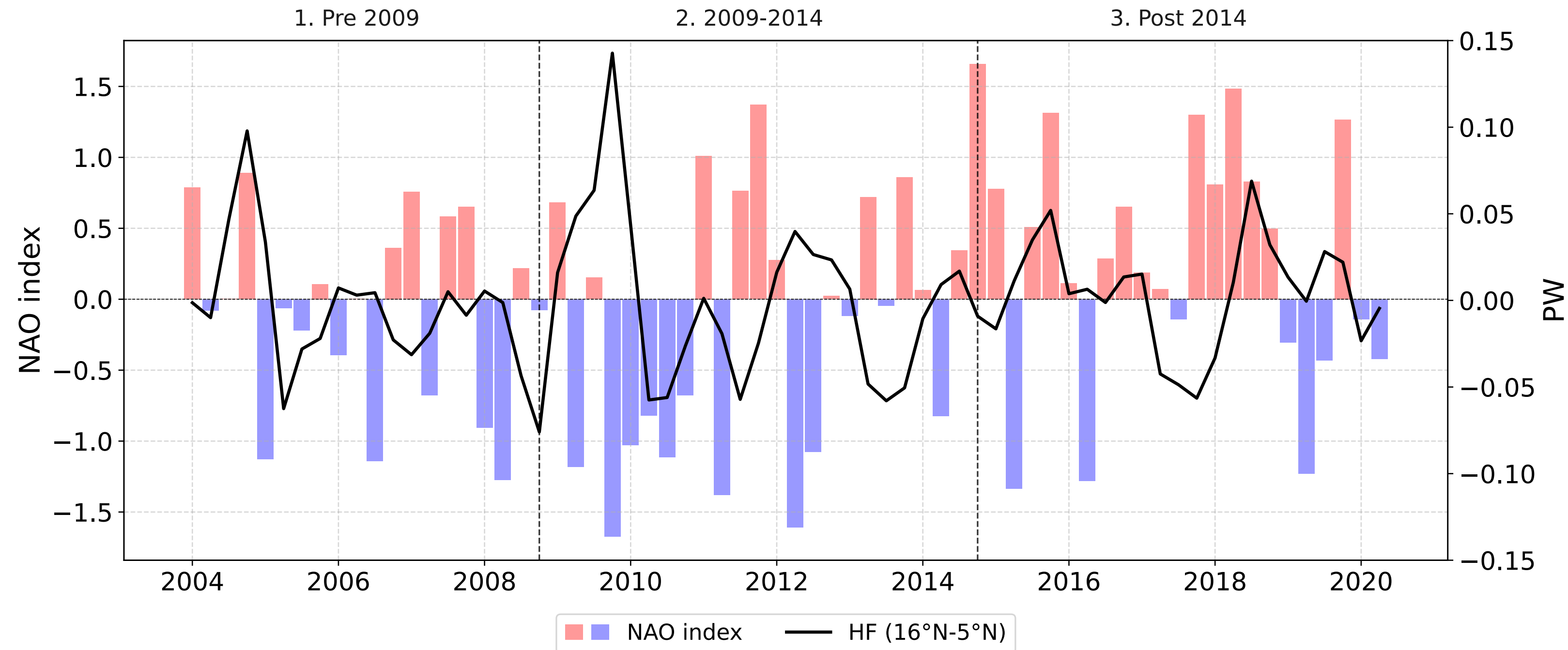
NAO & OHCT anomalies



Period	16°N-5°N
Pre 2009	0.03
2009-2014	-0.37
Post 2014	-0.20

Correlation of OHCT in region with NAO index for each period, blue shaded values are significant (95%)

NAO & HF anomalies



- 2009-2014: slightly anti-correlated NAO & HF → negative NAO coincide with positive heat uptake
- Very similar to NAO & OHCT correlations

Period	16°N-5°N
Pre 2009	0.02
2009-2014	-0.40
Post 2014	0.06

Correlation of HF in region with NAO index for each period, blue shaded values are significant (95%)